CURVE SKETCHING

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Name: Solutions

1. Sketch the curve

$$f(x) = 3x^4 - 8x^3 + 6x^2$$

(a) State the domain of f.

$(-\infty, \infty)$

(b) Find the intercepts and express them as an (x, y) pair. Write NONE if there are none.

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

$0 = f(x) = 3x^4 - 8x^3 + 6x^2 = x^2(3x^2 - 8x + 6)$

$D = (-8)^2 - 4(3)(6)$

$= 64 - 12(6) = 64 - 72 < 0 \Rightarrow 3x^2 - 8x + 6$ irreducible.

(c) Is the function even, odd, or neither? What type of symmetry does the function have?

$f(-x) = 3(-x)^4 - 8(-x)^3 + 6(-x)$

$= 3x^4 + 8x^3 - 6x \neq -f(x), f(x)$

So neither.

(d) Find the asymptotes. Write NONE if there are none.

Horizontal: NONE

Vertical: NONE

(e) Find the intervals where the function is increasing and decreasing. Write NONE if not applicable.

Increasing: $(0, \infty)$

Decreasing: $(-\infty, 0)$

(f) State the local maximum and local minimum value(s). Write NONE if not applicable.

Local maximum value(s): NONE

Local minimum value(s): $(0,0)$

(g) Find the intervals on which the function is concave up and concave down. State the inflection points. Write NONE if not applicable.

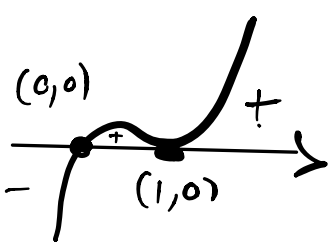
Concave Up: $(-\infty, \frac{1}{3}) \cup (1, \infty)$

Concave Down: $(1/3, 1)$

Inflection Points: $x=1, \ x = 1/3$

$f'(x) = 12x^3 - 24x^2 + 12x$
$= 12x(x^2 - 2x + 1)$
$= 12x(x-1)^2$

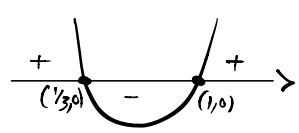
$\Leftrightarrow x = 0$ or $x = 1$



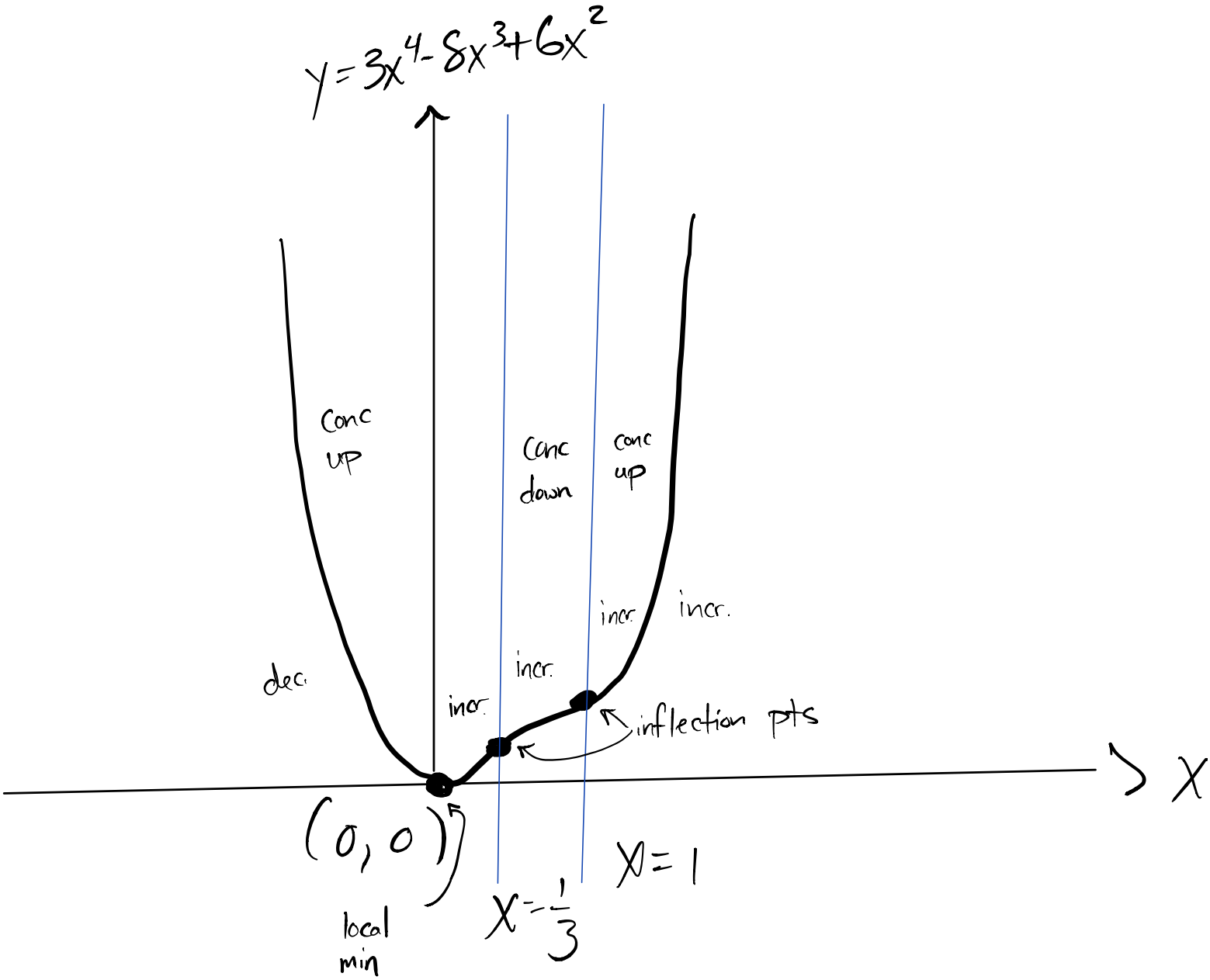
$x = 0$ min

$f''(x) = 36x^2 - 48x + 12$
$= 12(3x^2 - 4x + 1)$
$= 12(3x - 1)(x - 1)$
$= 0$
$\Leftrightarrow x = 1$ or $x = \frac{1}{3}$.

both inflection points.



(h) Use your answers to Parts (a)-(g) to sketch the curve. Be sure that your graph is labeled and neat.



2. Sketch the curve

$$f(x) = \frac{2x^2}{x^2 - 1}$$

(a) State the domain of f.

$x \neq \pm 1$.

(b) Find the intercepts and express them as an (x, y) pair. Write NONE if there are none.

x-intercept(s): $(0,0)$

y-intercept: $(0,0)$

(c) Is the function even, odd, or neither? What type of symmetry does the function have?

Even:

$$f(-x) = \frac{2(-x)^2}{(-x)^2-1} = \frac{2x^2}{x^2-1} = f(x)$$

Symmetry about y-axis.

(d) Find the asymptotes. Write NONE if there are none.

Horizontal: $y = 2$

Vertical: $x = \pm 1$

(e) Find the intervals where the function is increasing and decreasing. Write NONE if not applicable.

Increasing: $(-\infty, -1) \cup (-1, 0)$

Decreasing: $(0, 1) \cup (1, \infty)$

(f) State the local maximum and local minimum value(s). Write NONE if not applicable.

Local maximum value(s): $(0, 0)$

Local minimum value(s): NONE

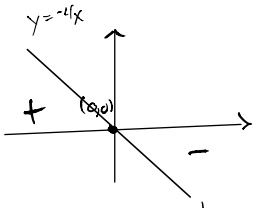
(g) Find the intervals on which the function is concave up and concave down. State the inflection points. Write NONE if not applicable.

Concave Up: $(-\infty, -1) \cup (1, \infty)$

Concave Down: $(-1, 1)$

Inflection Points: NONE ($x = \pm 1$ not in domain of f)

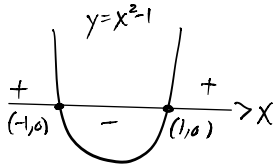
$$f'(x) = \frac{4x(x^2-1) - 2x^2(2x)}{(x^2-1)^2}$$
$$= \frac{4x^3 - 4x - 4x^3}{(x^2-1)^2}$$
$$= \frac{-4x}{(x^2-1)^2} = 0 \iff x = 0$$



$\Rightarrow f'(x) < 0$ when $x < 0$

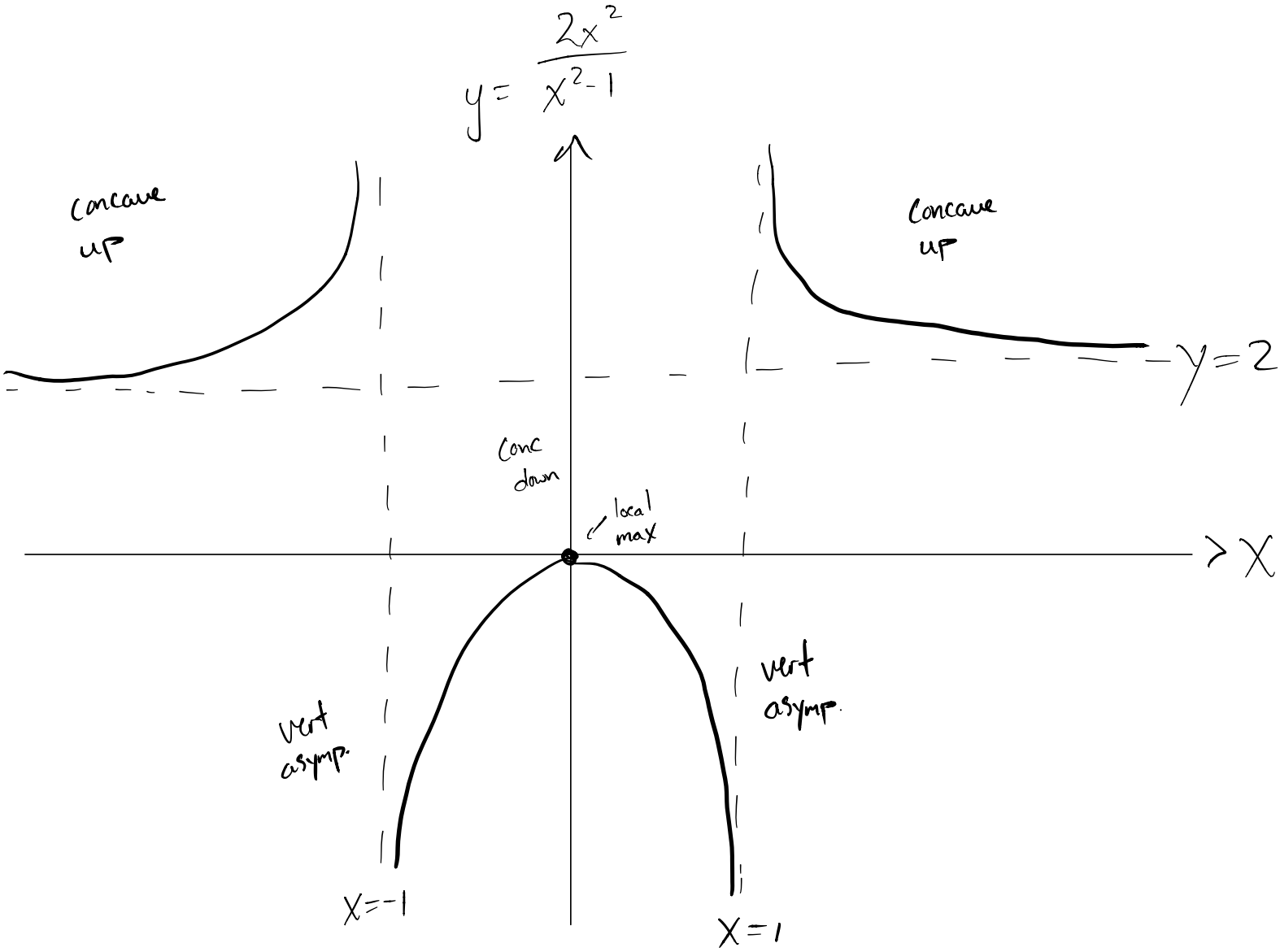
$f'(x) > 0$ when $x > 0$

$$f''(x) = \frac{-4(x^2-1)^2 + 4x(2)(x^2-1)(2x)}{(x^2-1)^4}$$
$$= \frac{(x^2-1)(-4(x^2-1) + 16x^2)}{(x^2-1)^4}$$
$$= \frac{-4x^2 + 4 + 16x^2}{(x^2-1)^3}$$
$$= \frac{12x^2 + 4}{(x^2-1)^3} \neq 0$$



$\Rightarrow f''(x) < 0$ on $(-1, 1)$

$f''(x) > 0$ else

$$\lim_{x \to \pm\infty} \frac{2x^2}{x^2-1} = \lim_{x \to \pm\infty} \frac{x^2}{x^2}\left(\frac{2}{1 - 1/x^2}\right)$$
$$= \lim_{x \to \pm\infty} \frac{2}{1 - \frac{1}{x^2}} = \frac{2}{1}$$

(h) Use your answers to Parts (a)-(g) to sketch the curve. Be sure that your graph is labeled and neat.